Chengyuan Xu

Adobe Inc., San Jose, California
chengyuanx@adobe.com
hello.xu@gmail.com
cy-xu.github.io

AREAS OF INTEREST

Multi-Modal Generative Models – video understanding, data moderation, ablation experiments & training, inference-time optimization, and scientific evaluation for video generation and video editing

Segmentation, Multi-object Tracking, Human-AI Collaboration, Interactive Machine Learning, Augmented Reality

EDUCATION

2017–24 Ph.D. in Media Arts and Technology, University of California, Santa Barbara
Committee: Tobias Höllerer, Jennifer Jacobs, Marko Peljhan, Curtis McCully

2020–24 M.S. in Computer Science, University of California, Santa Barbara

PROFESSIONAL EXPERIENCE

2023- **Adobe Firefly**
Research Engineer

2026 **Foundation model evaluation lead**

- Define scientific evaluation goals and interpreting signals for Firefly's multi-modal generative model throughout the production pre-training and post-training stages.
- Building agentic workflows for inference-time improvements e.g., agentic video quality judges and prompt rewriting optimization.

2025 **Large-scale video understanding and moderation pipeline**

- Co-designed Adobe Firefly's in-house video data moderation pipeline, reducing training data preparation time for 100 millions of videos from months to days.

2025 **Adobe Firefly video model major updates**

- Core contributor to data pre-processing, training ablation, application-specific mid-training, and SFT post-training experiments for Adobe's multi-modal foundation model.

2024 **Adobe Firefly video model public beta**

- Core member of Firefly's first public beta model training. Designed and trained the flickering-free video decoder to unblock product release.

Research Scientist/Engineer Intern, Summer Internship

2023 **Internship Model Shipped in Adobe Product**

- Trained a GAN-based facial editing model and released in Photoshop Express on iOS and Android.

2022 **Appen Limited**
Computer Vision Intern, Dec. 2021 – Sept. 2022

- We proposed a AI-in-the-loop video annotation workflow that helped human annotators track faces in very long videos 30% faster with better quality.
- We designed a large user study and investigated 3,466 person-hours of annotation work in "Comparing Zealous and Restrained AI Recommendations in High-stakes Human-AI Collaboration." The analysis revealed significant findings to guide future designs of human-AI collaboration systems in high-stakes tasks.

2021 **Benioff Ocean Science Laboratory**
Computer Vision Researcher, Summer Internship

- A trash detection model and a dataset "BOI Baltimore Trash Wheel Computer Vision Model and Dataset", deployed in Baltimore to identify 15 types of river waste. The work collects data and assist marine scientists in better understanding the types and sources of ocean-bound river waste.

2019 **Las Cumbres Observatory**
Imaging Intern, Summer Internship

- First author of top astrophysical journal paper "Cosmic-ConNN." The segmentation model was deployed to 20+ astronomical telescopes globally, processing thousands of space observations daily.

PUBLICATIONS

2025 Zhu, Z., Duarte, K., · Rizve, MN., **Xu, C.**, Kalarot, R., & Yuan, J."CompSlider: Compositional Slider for Disentangled Multiple-Attribute Image Generation." ***IEEE ICCV 2025*** *proceedings.*

2024 **Xu, C.**, Kumaran, R., Stier, N., Yu, K., & Höllerer, T. "Multimodal 3D Fusion and In-Situ Learning for Spatially Aware AI." ***IEEE ISMAR 2024*** *oral presentation.*

2023 Zhu, J., Kumaran, R., **Xu, C.**, & Höllerer, T. "Free-form Conversation with Human and Symbolic Avatars in Mixed Reality." ***IEEE ISMAR 2023*** *oral presentation*.

2023 **Xu, C.**, Lien, K.C., & Höllerer, T. "Comparing Zealous and Restrained AI Recommendations in High-stakes Human-AI Collaboration." ***ACM CHI 2023*** *oral presentation*.

2022 **Xu, C.**, Dong, B., Stier, N., McCully, C., Howell, D. A., Sen, P., & Höllerer, T. "Interactive Segmentation and Visualization for Tiny Objects in Multi-megapixel Images." ***CVPR 2022*** *demo* *and* *proceedings*.

2022 **Xu, C.**, McCully, C., Dong, B., Howell, D. A., & Sen, P. "Cosmic-CoNN: A Cosmic Ray Detection Deep Learning Framework, Dataset, and Toolbox."
240th American Astronomical Society meeting*, oral presentation.*
The Astrophysical Journal*, Volume 942, Number 2.*

2021 Hiramatsu, D. et al., including **Xu, C.** "The electron capture origin of supernova 2018zd."
Nature Astronomy*, cover story Volume 5, Issue 9.*

SERVICE

2025	Reviewer, IEEE ISMAR 2025, Experimental Astronomy Journal
2024	Reviewer, ACM CHI 2024, ACM IMX 2024, ACM VRST 2024, IEEE ISMAR 2024, IEEE TCSVT
2023	Reviewer, ACM CHI PLAY 2023, IEEE PacificVis 2024
2022	Reviewer, ACM CHI 2023
2021–22	Student Representative, Media Arts and Technology Program, UCSB.
2020–21	Peer Mentor, Women In Computer Science, WiCS Mentorship Program, UCSB.

ACADEMIC EXPERIENCE

2023-25	SIGCHI member, the ACM Special Interest Group on Computer-Human Interaction
2021	SDSC Cyberinfrastructure-Enabled Machine Learning Summer Institute
2018–23	Graduate Student Researcher, University of California, Santa Barbara

GRANTS AND AWARDS

2018-22	International Doctoral Recruitment Fellowship ($15,000 Annually).
2020	Mellichamp 21st Century Global Dynamics Graduate Research Fellowship ($7,500).
2018	Media Arts and Technology Grant ($2,500).
2018	MAT End of Year Show Grant ($750).